CHATTER: A Character Attribution Dataset for Narrative Understanding

Sabyasachee Baruah[1] **and Shrikanth Narayanan**[2]
Signal Analysis & Interpretation Laboratory
University of Southern California
sbaruah@usc.edu[1], shri@usc.edu[2]

Abstract

Computational narrative understanding studies the identification, description, and interaction of the elements of a narrative: characters, attributes, events, and relations. Narrative research has given considerable attention to defining and classifying character types. However, these character-type taxonomies do not generalize well because they are small, too simple, or specific to a domain. We require robust and reliable benchmarks to test whether narrative models truly understand the nuances of the character's development in the story. Our work addresses this by curating the CHATTER dataset that labels whether a character portrays some attribute for 88124 character-attribute pairs, encompassing 2998 characters, 12967 attributes and 660 movies. We validate a subset of CHATTER, called CHATTEREVAL, using human annotations to serve as a benchmark to evaluate the character attribution task in movie scripts. CHATTEREVAL also assesses narrative understanding and the long-context modeling capacity of language models.

1 Introduction

Narrativity occurs when characters interact with each other, triggering events that are temporally, spatially, and causally connected. This sequence of events forms the story. Piper et al. (2021) provided a symbolic definition of narrativity in which they asserted that narrativity occurs when the narrator $\mathcal{A}$ tells the perceiver $\mathcal{B}$ that some agent $\mathcal{C}$ performed the action $\mathcal{D}$ on another agent $\mathcal{E}$ at place $\mathcal{F}$ and time $\mathcal{G}$ for some reason $\mathcal{H}$. Baruah and Narayanan (2024) used this definition to identify four main elements of any narrative: characters, attributes, events, and relations. Labatut and Bost (2019) explored different types of character interactions and emphasized the central role characters play in narratives. Characters drive the plot forward through their actions, develop attributes, arouse tension and emotion in the story by creating conflicts or bonds with other characters, and embody tropes and stereotypes to relate to the audience. The vitality of characters in narratives makes character understanding an essential task in narrative research.

Dataset	Domain	Type	Size
Finlayson (2015)	Folklore	Archetype	282
Skowron et al. (2016)	Movies	Role	2010
Brahman et al. (2021)	Literature	Description	9499
Sang et al. (2022)	Movies	Personality	28653
Yu et al. (2023)	Literature	Traits	52002
CHATTER	Movies	Tropes	88124
CHATTEREVAL	Movies	Tropes	1061

Table 1: Comparison with other attribution datasets in terms of the attribute type and size. Size denotes the number of character-attribute pairs. CHATTER is the largest character attribution dataset collected so far.

Narratologists have explored various approaches to operationalize the character-understanding task. For example, Inoue et al. (2022) presented character understanding as a suite of document-level tasks that included gender and role identification of the character, *cloze* tasks, quote attribution, and question answering. Li et al. (2023) adopted coreference resolution, character linking, and speaker guessing tasks, and Azab et al. (2019) used character relationships and relatedness to evaluate character representations. We organized the character-understanding tasks into the following categories. **1) Identification** tasks find the unique set of characters and their mentions. It includes entity recognition, entity linking, and coreference resolution. The **2) Quotation** task maps utterances to characters. **3) Attribution** tasks, such as personality classification, persona modeling, and description generation, describe the character. The **4)** ***Cloze*** task asks the model to fill in the correct character name given an anonymized character description, story summary, or story excerpt. **5) Relation** tasks, such as relation classification, draw similarities between characters.

Among these tasks, character attribution is the most challenging because there exists a multitude of ways to qualify a character, such as personality (Sang et al., 2022), adjectives (Yu et al., 2023), persona (Bamman et al., 2013, 2014), archetypes (Finlayson, 2015), roles (Skowron et al., 2016), and descriptions (Brahman et al., 2021). Each of these approaches has drawbacks. For example, personality scales such as Big5 and MBTI cannot capture all the variation in characterization and can be difficult to interpret (Zillig et al., 2002). Adjective descriptors are too general and apply only to a limited context in the story. Persona roles are not explicitly defined. Archetypes (Propp, 1968; Jung, 2014) are abstract concepts specific to the domain of interest. Character descriptions are detailed, freeform, and scalable to any number of characters. However, we cannot factor them into simpler components or use them to compare between different characters.

We require a robust attribution taxonomy that can scale across different narratives, characters, and domains, is well-defined and discrete for effective character comparison, and necessitates document-level understanding to model them accurately. We developed the CHATTER dataset to fulfill this need. The CHATTER dataset uses tropes from the TVTropes website as the attribute type to describe characters. TVTropes editors and moderators comprehensively define every trope with illustrative examples from multiple media sources. Tropes cover a wider range of character descriptions than personality types and archetypes, and require longer context-understanding, compared to traits and adjectives, to accurately ascertain if a character portrays some trope. Unlike character descriptions, we can efficiently compare characters and their experiences using these well-defined tropes (Wang et al., 2021).

The CHATTER dataset contains labels indicating whether the character portrayed the trope in the movies they appeared in. It contains 88124 character-trope pairs. We drew our characters primarily from Hollywood movies across various genres. It also provides the full-length screenplays of the movies, averaging about 25K words per screenplay. We define the character attribution task as a **binary classification task** where, given a character-trope pair and the screenplays of the movies where the character appears, the model should predict whether the character portrayed the trope. We validate a subset of the CHATTER data, referred to as CHATTEREVAL, using human annotations to establish an evaluation benchmark for the character attribution task. We compared the zero-shot and few-shot performance of LLMs and CHATTER's labels to assess the suitability of using the CHATTER dataset as a training set for the attribution task. The dataset is available at https://drive.google.com/drive/folders/11egMhs-zkWSASe7zJENwHg17-6VOeXDU?usp=sharing.

2 Data

2.1 Tropes

Tropes are storytelling devices or conventions used by the writer to easily convey some story notion to the reader. They act like narrative motifs that readers can easily recognize, saving time and effort for the writer as they can omit details the readers can infer from the trope. For example, the *AntiHero* is a very popular trope that describes a protagonist who lacks traditional heroic qualities, often cynical and flawed, yet ultimately performs heroic actions. The character Severus Snape in the Harry Potter stories portrays the *AntiHero* trope as he gives Harry a hard time but stays loyal to Dumbledore (mentor) and works secretly to defeat Voldemort (antagonist).

Tropes can also relate to inanimate objects, events, locations or the environment. We focus only on character tropes in our work. The scope of a trope can vary greatly. Some tropes like *PetTheDog* (villain performing an act of kindness) are portrayed over short contexts, typically a single scene, whereas others like *HiddenDepths* (revealing unexpected talents as the story progresses) are portrayed over a longer context. The attribution model should be able to extract information from different points in the narrative and reason over it to identify the portrayed tropes.

We use the character trope labels of TVTropes[1]. TVTropes is a community-driven website, similar to Wikipedia, that catalogs tropes with definitions and examples. Fans of a creative work discuss and post tropes they identify in the narrative. Each trope has a dedicated page which contains its definition, illustrations, and portrayal examples from TV shows, movies, literature, animation, video games, and print media. TVTropes moderators ensure that the fan-edited content is correct. We collect the trope annotations from TVTropes to build the CHATTER dataset.

It is important to note that the character trope annotations we collect from TVTropes are those *perceived* by the reader. These might not align with

[1] https://tvtropes.org

the *actual* tropes intended by the creator. Since there is no quantifiable agreement on the published content, we treat the TVTropes data as a noisy source of character attribution.

2.2 Screenplays

We used movies as the source of our narratives. We chose the cinematic domain over the literary domain because it had more TVTropes labels, and we supposed it would be easier to find raters more knowledgeable about movies than books. Additionally, movies allow us to extend the attribution task to the multimodal domain, offering more opportunities for future research directions.

We used publicly available movie screenplays from the ScriptsonScreen[2] website. Each script is mapped to an IMDB[3] identifier so we can uniquely identify the movie. Most movies in our dataset are produced in the US or the UK after 1980. The average script size is about 25K words. We apply a named entity classifier and name alias generator to map the character names in the script to a unique character in the IMDB cast list. We preprocess the screenplays to find scenes, dialogues and descriptions using Baruah and Narayanan's (2023) screenplay parser. In total, our dataset contains screenplays of 660 movies.

2.3 CHATTER

We build the CHATTER dataset of character-trope pairs using the tropes of TVTropes and the screenplays of ScriptsonScreen. First, we download the movie screenplays from ScriptsonScreen, parse and map them to an IMDB page, and match the characters occurring in the document to a character in the IMDB cast list. Second, we search for the character in TVTropes and retrieve their character page. Third, we collect the tropes portrayed by the character from the character page, as labeled by the TVTropes community. Fourth, we search for the trope page and fetch its definition. We also summarize the definition using GPT-4 for the annotation task. The average size of the trope definition and its summary is 344 and 44 words, respectively.

We need good negative samples to evaluate the specificity of the attribution model. However, TVTropes does not provide this information because it does not label tropes *not* portrayed by the character. Instead, we analyze the trope definition to find antonym tropes and create the negative character-trope pairs. For example, the definition of the *AntiHero* trope contains the sentence –

`...Compare and contrast this trope with its antithesis, the AntiVillain...`

– which indicates that the *AntiVillain* trope is contrary to *AntiHero*. We search the trope definitions to retrieve opposing tropes by checking if negation words such as "contrast", "opposite" and "counterpart" exist within a five-word context window of the antonym trope mention[4]. For each positive character-trope pair, we create a negative pair by either – 1) selecting an antonym trope from the trope definition or 2) choosing a trope that is absent in any of the positive character-trope pairs for that character. It should be harder for the attribution model to distinguish antonym tropes from tropes portrayed by the character because they are much more closely related to the portrayed tropes than a trope randomly sampled with the second method. Therefore, the former method creates hard negatives and the latter method creates soft negatives. We randomly choose between the two methods with 50% probability to get a good mix of hard and soft negatives. We add the negative samples to complete the construction of the CHATTER dataset.

3 Evaluation Data

We can use CHATTER to train attribution models, but we require a more reliable dataset for evaluation. We annotate a subset of CHATTER using human raters and create the CHATTEREVAL dataset.

3.1 Annotation

We sampled movies from our dataset released after 2010 and collected the corresponding character-trope pairs for annotation. We employed workers on the Amazon MTurk[5] crowdsource annotation platform. We selected workers with high reputation and experience[6] and tested them on two separate qualification tasks, each containing five questions that asked them to decide whether the character portrayed the given trope. The task showed them the picture of the character, the movies that starred the character with Wikipedia links, a link to the fandom page[7] where the worker can read more about the character, and the summarized trope definition with a link to the trope's TVTropes page. Appendix A

[2] https://scripts-onscreen.com

[3] https://imdb.com

[4] The full list of negation words is included in the dataset

[5] https://www.mturk.com

[6] >98% approval rate and worked on >5000 HITs

[7] https://www.fandom.com